



WORK SHEET - 01

VECTOR & 3D

[HINT & SOLUTION]

Q.1 (C)

Q.2 (C)

Q.3 (B)

Hint : $2x + y + 4z - 5 = 0$

Q.4 (A)

Q.5 AB

Sol. Let vector parallel to plane is $\vec{V} = a\hat{i} + b\hat{j} + c\hat{k}$
as plane is parallel to $l : x = 0, y = -z$ i.e.,

$$l : \frac{x}{0} = \frac{y}{1} = \frac{z}{-1}$$

$$\Rightarrow \vec{V} \cdot (\hat{j} - \hat{k}) = 0$$

$$b = c$$

$$\vec{V} = a\hat{i} + b\hat{j} + b\hat{k}$$

Now, angle of plane with l_1 is $\pi/4$

$$l_1 : \frac{x}{1} = \frac{y}{1} = \frac{z}{0}$$

$$\vec{V} \wedge l_1 = \frac{\pi}{4} \Rightarrow \cos\left(\frac{\pi}{4}\right) = \vec{V} \cdot (\hat{i} + \hat{j})$$

$$\frac{a+b}{\sqrt{2} \cdot \sqrt{a^2+2b^2}} = \frac{1}{\sqrt{2}}$$

$$\Rightarrow a^2 + b^2 + 2ab = a^2 + 2b^2$$

$$\Rightarrow b^2 = 2ab \text{ if } b \neq 0$$

$$\Rightarrow 2a = b$$

Required equation

If $O = b \neq 0$

$ax + 2ay + 2az = 5a$ as passes $(-1, 2, 1)$

$$\Rightarrow x + 2y + 2z = 5$$

and if $b = 0$

$$ax + by + bz = -a + 2b + b$$

$$\Rightarrow ax = -a \text{ as } P \text{ contains } (-1, 2, 1)$$

$$\Rightarrow x = -1 \Rightarrow x + 1 = 0$$



Thus required equation of plane is

$$P: x + 2y + 2z = 5 \text{ or}$$

$$P: x + 1 = 0$$

Q.6 AB

Sol. $l_1: x + y = 2$

$$x = y + z$$

$$l_2: \frac{x}{1} = \frac{y-2}{-1} = \frac{z+2}{2} = \lambda$$

$$l_2: \frac{x}{1} = \frac{y}{1} = \frac{z+1}{2} = \mu \text{ (Given)}$$

Point of intersection of l_1 and l_2

$$x, \lambda = \mu$$

$$y = \mu = 2 - \lambda$$

$$z = \mu - 1 = 2\lambda - 2$$

$$\lambda = 1 \text{ and } \mu = 1$$

$\Rightarrow$ Point of intersection is $(1, 1, 0)$

Angle between l_1 and l_2

$$l_1 \wedge l_2 = \cos^{-1} \left(\frac{2}{\sqrt{3}\sqrt{6}} \right) = \sin^{-1} \left(0, \frac{\sqrt{7}}{3} \right)$$

Any point on l_1 can be taken as $(\lambda, 2 - \lambda, 2\lambda - 2)$

$$d \times \frac{\sqrt{7}}{3} = \frac{\sqrt{14}}{3}$$

$$\Rightarrow d = \sqrt{6}$$

$$\Rightarrow (\lambda - 1)^2 + (1 - \lambda)^2 + (2\lambda - 2)^2 = 6$$

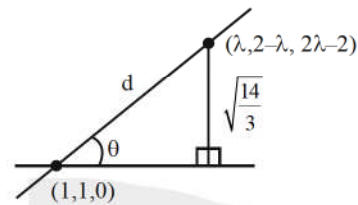
$$\Rightarrow (\lambda - 1)^2 = 1$$

$$\Rightarrow \lambda - 1 = \pm 1$$

$$\lambda = 0 \text{ or } \lambda = 2$$

Points are

$$(0, 2, -2) \text{ and } (2, 0, 2)$$



[PARAGRAPH TYPE]

Q.7 $x^2 + y^2 = \frac{r^2}{9}$ and 7.5

Sol. $A = (x_1, y_1)$

$$B = (x_1, 0)$$

$$C = (-x_1, y_1)$$

$$BC = 2y_1x + x_1y = x_1y_1$$

$$AO = y_1x - x_1y = 0$$

$$\text{intersect at } \left(\frac{x_1}{3}, \frac{y_1}{3} \right)$$



The locus of points is a circle with $\frac{1}{3}$ the radius.

$$x^2 + y^2 = \frac{R^2}{9}$$

[NUMERICAL INTEGER TYPE]

Q.8 $\frac{1}{\sqrt{52}}$

Hint. Locus is $6x + 4y = 1$ & $\frac{1}{\sqrt{52}}$ is minimum distance.

Sol. as given AB, AP and AC are in harmonic progression

Using parametric

Any point on the curve $2x^2 + y^2 = 2$

or $x^2 + \frac{y^2}{2} = 1$ can be taken as

$$(\cos \theta, \sqrt{2} \sin \theta)$$

Using harmonic conjugate as B and C are harmonic conjugate of each other.

$$\cos \theta = \frac{\lambda h + 6}{\lambda + 1} \text{ and } \sqrt{2} \sin \theta = \frac{\lambda k + 6}{\lambda + 1}$$

$$\cos \phi = \frac{\lambda h - 6}{\lambda - 1} \text{ and } \sqrt{2} \sin \phi = \frac{\lambda k - 6}{\lambda - 1}$$

using $\sin^2 A + \cos^2 A = 1$

$$\left(\frac{\lambda h + 6}{\lambda + 1} \right)^2 + \frac{1}{2} \left(\frac{\lambda k + 6}{\lambda + 1} \right)^2 = 1 \quad \dots(i)$$

$$\left(\frac{\lambda h - 6}{\lambda - 1} \right)^2 + \frac{1}{2} \left(\frac{\lambda k - 6}{\lambda - 1} \right)^2 = 1 \quad \dots(ii)$$

Subtracting (ii) from (i) we get

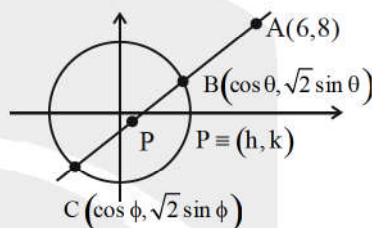
$$2(2\lambda h)(12) + (2\lambda k)(16) = 8\lambda$$

$$6h + 2k = 1$$

Required locus is $6h + 2k = 1$

Minimum distance from origin

$$\left| \frac{1}{\sqrt{36 + 16}} \right|, \frac{1}{\sqrt{152}}$$



Q.9 [1]

Q.10 [6]

Sol. First we see that $f(R) = 4$ since it lies on line $ax + by = 4$.

Second, we see that f is linear, but we need only additivity. Take any $A(x, y)$ and $B(x', y')$:

$$f(A + B) = a(x + x') + b(y + y') = ax + by + ax' + by' = f(A) + f(B)$$

Since $\overrightarrow{P_2R} = 2\overrightarrow{RP_1}$ we have $R - P_2 = 2(P_1 - R)$ thus $3R = 2P_1 + P_2$.

Since $3R = 2P_1 + P_2$ and f is linear function we have

$$f(P_2) = f(3R - 2P_1) = 3f(R) - 2f(P_1) = 3 \cdot 4 - 2 \cdot 3 = 6$$

Q.11 [6]

Q.12 [16]

Q.13 $4, \frac{3}{11}$

Q.14 $9x + 7y - z = 4$

Q.15 $\frac{2}{\sqrt{6}}$

Q.16 $\frac{2}{3}$

Sol. Using Vol. (OABP) + Vol. (OACP) + Vol. (ORBC) + Vol. (PABC) + Vol. (OABC)
Volume of OABC i.e. Vol. (OABC) is

$$\left| \frac{1}{6} [\vec{a} \ \vec{b} \ \vec{c}] \right| = \left| \frac{1}{6} \begin{vmatrix} 0 & 0 & 2 \\ 0 & 1 & 0 \\ 6 & 0 & 0 \end{vmatrix} \right|$$

$$= \frac{1}{6} \times 24 \times 2 = 8$$

$$\text{Vol. (OABP)} = \frac{1}{6} \cdot r \cdot 8$$

$$\text{Vol. (OACP)} = \frac{1}{6} \cdot r \cdot 24$$

$$\text{Vol. (OPBC)} = \frac{1}{6} \cdot r \cdot 12$$

$$\text{Vol. (PABC)} = \frac{1}{6} \cdot r \cdot 28$$

$$6r + 6r = 8$$

$$r = \frac{2}{3}$$



Q.17 $\vec{r} = (1, 2, 0) + \lambda (2, -2, 1)$ or

$$\vec{r} = \left(1, \frac{10}{9}, \frac{-8}{9}\right) + \lambda (12, -12, 21)$$

Q.18 [3]

Rankers Academy JEE

